

Analysis I

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Lecture 1 – 22/01/26

Analysis is about things that are small, things that are large, and quantifyig how small or large they are.

§1 Numerical sequences and series

§1.1 Basics

Definition 1.1. A sequence $(x_n)_{n \in \mathbb{N}}$ on a set X is an enumerated list where each element is in X .

In this chapter, $X \subseteq \mathbb{R}$ or $X \subseteq \mathbb{C}$.

What does it mean for a sequence (x_n) to converge to a point $x \in X$? What does it mean for (x_n) to diverge to infinity?

For convergence, we want $|x_n - x|$ to be small. But what is small? In fact, we want $|x_n - x|$ to be arbitrarily small, i.e. smaller than any $\varepsilon > 0$ we choose. What really matters is the *tail* of the sequence – we can always ignore the first N terms, and we should pick N depending on ε .

For divergence to infinity, we want $|x_n|$ to get arbitrarily large, i.e. larger than any $L > 0$ we choose. Again, we ignore the first N terms depending on L .

Definition 1.2. We say (x_n) converges to some finite x , $x_n \rightarrow x$, if, for any $\varepsilon > 0$, there exists $N(\varepsilon)$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$.

We say (x_n) diverges to infinity if, for any $L > 0$, there exists $N(L)$ such that $x_n > L$ for all $n \geq N$, analogously for negative infinity. For complex sequences, we must consider $|x_n|$.

- Example 1.3**
- Take $x_n = \frac{1}{n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{n} < \varepsilon$ for any $n \geq N = 1 + \lceil \frac{1}{\varepsilon} \rceil$.
 - Take $x_n = \frac{1}{2^n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{2^n} < \varepsilon$ for any $n \geq N = \max(1 + \lceil \log_2(\frac{1}{\varepsilon}) \rceil, 1)$.
 - Take $x_n = in$, then $x_n \rightarrow \infty$, since $\frac{x_n}{i} = n > L$ for any $n \geq N = 1 + \lceil L \rceil$.
 - Take $x_n = (-1)^n$, then x_n doesn't converge or diverge to infinity.

Lemma 1.4

If it exists, the limit of (x_n) is unique.

Proof. Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Take $\varepsilon > 0$, then we have

$$\begin{aligned} \exists N_1 = N_1(\varepsilon) \text{ s.t. } |x_n - a| < \varepsilon \quad \forall n \geq N_1 \\ \exists N_2 = N_2(\varepsilon) \text{ s.t. } |x_n - b| < \varepsilon \quad \forall n \geq N_2 \end{aligned}$$

Now take $N = \max(N_1, N_2)$. We have

$$\begin{aligned} |a - b| &= |a - x_n + x_n - b| \\ &\leq |x_n - a| + |x_n - b| \\ &< 2\varepsilon \end{aligned}$$

for all $\varepsilon > 0$. It follows that $|a - b| = 0$, so $a = b$. ■

This justifies the notation $\lim_{n \rightarrow \infty} x_n = x$.

Proposition 1.5 (Sandwich theorems)

Let $(x_n), (y_n), (z_n)$ be real sequences. Then

- If $x_n \leq y$ for all n and $x_n \rightarrow x$, then $x \leq y$. Note that $x_n < y$ does not imply $x < y$.
- If $x_n \rightarrow x$, $z_n \rightarrow x$ and $x_n \leq y_n \leq z_n$ for all n , then $y_n \rightarrow x$.

Proof. Left as an exercise. ■

Lemma 1.6

Let (x_n) be a complex sequence. Then $x_n \rightarrow x$ if and only if $\operatorname{Re}(x_n) \rightarrow \operatorname{Re}(x)$ and $\operatorname{Im}(x_n) \rightarrow \operatorname{Im}(x)$.

Proof. Recall that, for $z \in \mathbb{C}$, $|z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$.

$\implies$ Use $|\operatorname{Re}(z)| \leq |z|$ and $|\operatorname{Im}(z)| \leq |z|$ with $z = x_n - x$.

$\Leftarrow$ Use $|z| \leq |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$, again with $z = x_n - x$. Fixing $\varepsilon > 0$, we can get $|x_n - x| < 2\varepsilon$, so we're done. ■

Lemma 1.7

Let $x_n \rightarrow x$, $y_n \rightarrow y$. Then $x_n + y_n \rightarrow x + y$, $x_n y_n \rightarrow xy$.

Proof. The first part is analogous to the above result on complex sequences and is an exercise.

For the second part, we have

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x y_n + x y_n - xy| \\ &\leq |y_n| |x_n - x| + |x| |y_n - y| \end{aligned}$$

Now, for $\varepsilon > 0$, we can pick an N to make $|x_n - x| < \varepsilon$, $|y_n - y| < \varepsilon$. Hence

$$\begin{aligned} |x_n y_n - xy| &< \varepsilon(|x| + |y_n|) \\ &< \varepsilon(|x| + Y) \end{aligned} \quad \text{by lemma 1.9, } Y > 0$$

and so picking $\varepsilon = \frac{\varepsilon'}{|x|+Y}$ shows the result. ■

Definition 1.8 (Bounded sequences). We say (x_n) is *bounded* if there exists $M > 0$ such that $|x_n| \leq M$ for all n , or, equivalently, if $\sup |x_n| \leq M$.

Lemma 1.9

Convergent sequences are bounded.

Proof. Take $\varepsilon = 1$, then there exists N such that $|x_n - x| \leq 1$ for any $n \geq N$, and so $|x_n| \leq 1 + |x|$ for any $n \geq N$. But for $n < N$, $|x_n| \leq \max_{1 \leq i < N} |x_i| < \infty$. ■

Definition 1.10 (Monotone sequences). We say a real sequence (x_n) is *monotone* if either

- it is increasing;
- or it is decreasing.

Proposition 1.11 (Monotone convergence theorem)

Every bounded monotone sequence converges.

Proof sketch. Without loss of generality, suppose the sequence is strictly increasing. Using the least upper bound property, we can show that $x_n \rightarrow \sup\{x_n\}$. ■

Can we strengthen to non-monotonic sequences? Clearly not fully, since $x_n = (-1)^n$ is bounded but not convergent. But, if we delete all the even terms, then it indeed becomes convergent.

§1.2 Bolzano-Weierstrass theorem

Theorem 1.12 (Bolzano-Weierstrass)

If (x_n) is a real bounded sequence, then it has at least one convergent subsequence.

Definition 1.13 (Subsequences). Given a sequence (x_n) , a subsequence of (x_n) is a sequence of the form $(x_{n_k})_{k \in \mathbb{N}}$, where (n_k) is a strictly increasing sequence of natural numbers.

Lemma 1.14

If $x_n \rightarrow x$, then any subsequence (x_{n_k}) of (x_n) must converge to x .

Proof. Since $n_k < n_{k+1}$, $n_{k+1} \geq n_k + 1$. By induction, we can show $n_k \geq k$.

Now, take $\varepsilon > 0$, so we have $N(\varepsilon)$ such that $|x_n - x| < \varepsilon$ for any $n \geq N$. So, if $k \geq N$, then $n_k \geq k \geq N$, and so $|x_{n_k} - x| < \varepsilon$, so $\lim_{k \rightarrow \infty} x_{n_k} = x$ as required. ■

Proposition 1.15 (Nested interval property)

Let $(I_n)_{n \in \mathbb{N}}$ be a sequence of closed intervals in $\mathbb{R}$ satisfying $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, and suppose $I_n = [a_n, b_n]$. If $b_n - a_n = |I_n| \rightarrow 0$ as $n \rightarrow \infty$, then $\bigcap_{n \in \mathbb{N}} I_n$ contains exactly one point.

Proof. We can see that (a_n) and (b_n) are bounded monotone sequences, so we probably want to apply [proposition 1.11](#). Indeed, we have $a_n \rightarrow a$, $b_n \rightarrow b$. Since limits preserve inequalities and $a_n \leq b_n$, we have $a \leq b$. Now we are ready to show the result.

- *Existence.* For all $k \geq n$, $a_k \in [a_k, b_k] \subseteq [a_n, b_n]$, and so $a_n \leq a_k \leq b_n$. Taking k to infinity, we have $a_n \leq a \leq b_n$ for every n , so $a \in I_n$ for every n , so it is in their intersection.
- *Uniqueness.* We know $b_n - a_n \rightarrow b - a$. But we know $b_n - a_n \rightarrow 0$, so $a = b$. Hence, if x is in the intersection, $a_n \leq x \leq b_n$ for all n , so $a \leq x \leq a$, and hence $x = a$ as required. ■

Proof of theorem. We are given (x_n) and $M > 0$ such that $|x_n| \leq M$ for any n . We will construct a series of nested intervals from which we can sample our subsequence and apply [proposition 1.15](#).

Let $a_1 = -M$, $b_1 = M$, so $I_1 = [-M, M] \supset (x_n)$. Now take $c = \frac{a_1 + b_1}{2}$. Then, at least one of the halves $[a_1, c]$ and $[c, b_1]$ will have infinitely many elements of (x_n) – we take I_2 to be a half that has infinitely many elements. Continue inductively to get (I_n) , which are nested by construction and all of which have infinitely many elements of the sequence – and by [proposition 1.15](#), there exists exactly one x in their intersection.

We choose (x_{n_k}) as follows: pick n_1 such that $x_{n_1} \in I_1$. Then I_2 has infinitely many elements of (x_n) with index greater than n_1 , so pick $n_2 > n_1$ such that $x_{n_2} \in I_2$. Continue inductively.

We have $x_{n_k} \in I_k$ for every k , so $x_{n_k} \in \bigcap_{n \leq k} I_n$, and so $x_{n_k} \rightarrow x$. ■

Corollary 1.16

The Bolzano-Weierstrass theorem holds in $\mathbb{C}$.

§1.3 Cauchy sequences

A Cauchy sequence is a sequence in which the elements are getting closer as the sequence progresses.

Definition 1.17. A sequence (x_n) on $X \subseteq \mathbb{C}$ is *Cauchy* if, for any $\varepsilon > 0$, there exists N such that, for all $n, m \geq N$, $|x_n - x_m| < \varepsilon$.

Example • Take $x_n = \frac{1}{n}$. Assume wlog $m \geq n$. Then

$$\left| \frac{1}{n} - \frac{1}{m} \right| = \frac{1}{n} - \frac{1}{m} = \frac{m - n}{mn} \leq \frac{1}{n} < \varepsilon$$

and this holds for any $m \geq n \geq N = \lceil \frac{1}{\varepsilon} \rceil + 1$. So (x_n) is Cauchy.

- Take $x_n = (-1)^n$. This is not a Cauchy sequence – take $n = 2k$, $m = 2k + 1$ for any $k \in \mathbb{N}$, so $|x_n - x_m| = 2$. Hence we can't take $\varepsilon < 2$.
- Take (x_n) on $\mathbb{Q}$ defined by truncation of the decimal expansion of $\sqrt{2}$. Assume wlog $m \geq n$, so

$$|x_n - x_m| < 10^{-(n-1)} \rightarrow 0$$

Note that this sequence does not converge in $\mathbb{Q}$.

Lemma 1.18

If (x_n) is Cauchy, then it is bounded.

Proof. Take $\varepsilon = 1$. Then we have N such that

$$|x_n - x_N| < 1$$

for any $n \geq N$ (we have fixed $m = N$). So $|x_n| < 1 + |x_N|$. ■

Lemma 1.19

A complex sequence (x_n) is Cauchy if and only if $(\operatorname{Re} x_n)$ and $(\operatorname{Im} x_n)$ are both Cauchy.

Lemma 1.20

If $x_n \rightarrow x$, then (x_n) is Cauchy.

Proof. For any $\varepsilon > 0$, we have N such that

$$\begin{aligned} |x_n - x_m| &\leq |x_n - x| + |x_m - x| \\ &\leq 2\varepsilon \end{aligned}$$

for $n, m \geq N$. This completes the proof. $\blacksquare$

Is the converse true? Not in $\mathbb{Q}$ at least – see $\sqrt{2}$ example. But we can still aspire to the converse being true in $\mathbb{R}$.

Theorem 1.21 (Completeness of $\mathbb{R}$ and $\mathbb{C}$)

Every Cauchy sequence in $\mathbb{R}$ or $\mathbb{C}$ converges.

Proof. Suppose (x_n) is Cauchy. Then it is bounded, and so by Bolzano-Weierstrass it has a convergent subsequence (x_{n_k}) converging to some x . We claim $x_n \rightarrow x$. Observe

$$|x_n - x| \leq |x_n - x_{n_k}| + |x_{n_k} - x|$$

Taking $\varepsilon > 0$, Cauchy gives $|x_n - x_{n_k}| < \varepsilon$ for $n, n_k \geq N_1$, and $x_{n_k} \rightarrow x$ gives $|x_{n_k} - x| < \varepsilon$ for $k \geq N_2$. Fixing $k \geq N_2$ such that $n_k \geq N_1$, we have $|x_n - x| < 2\varepsilon$ for every $n \geq N_1$, which suffices for the result. $\blacksquare$

Remark. This is nice – it's our first way to prove convergence without having to guess the limit.

§1.4 Series and convergent series

Definition 1.22. Let (a_n) be a sequence in $\mathbb{R}$ or $\mathbb{C}$. We say that $\sum_{n=1}^{\infty} a_n$ is a *series*, and we say it converges if the sequence of partial sums $s_k = \sum_{n=1}^k a_n$ converges to some finite s . In this case, s is called the sum of the series, and we write

$$s = \sum_{n=1}^{\infty} a_n$$

Example • $\sum_{n=1}^{\infty} n$ does not converge: $s_k = \frac{n(n+1)}{2}$.

• geometric series: $\sum_{n=1}^{\infty} r^n$ converges iff $|r| < 1$. We have $s_k = \frac{1-r^{k+1}}{1-r}$ (unless $r = 1$).